PulsePatch

Product Requirements Document (PRD)

Stick. Detect. Guide. Save.

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Product Stage	Concept to MVP Planning
Intended Audience	Product, Engineering, Clinical Advisors, Partners, Investors
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Purpose
This document defines the business rationale, product requirements, and execution roadmap for PulsePatch: a zero-friction emergency response patch designed to help bystanders deliver timely, effective CPR during suspected cardiac arrest events.

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1 Executive Summary

PulsePatch is a stick-on emergency response patch intended for rapid deployment during suspected sudden cardiac arrest (SCA). The product combines basic physiological sensing and motion analysis with real-time CPR guidance, helping untrained or minimally trained bystanders initiate and maintain higher-quality compressions until EMS arrives or an AED is deployed.

PulsePatch is positioned as a **non-diagnostic, intervention-guidance aid** designed to reduce hesitation and improve CPR quality in high-stress scenarios.

2 Problem Statement

Outcomes in sudden cardiac arrest are highly time-sensitive. Survival probability decreases significantly with each minute of delay before effective CPR.

Observed barriers in real-world response:

- Bystanders hesitate due to panic, uncertainty, or fear of doing harm.
- CPR quality is often inconsistent (depth, rhythm, continuity).
- AED availability is not universal in the first minutes after collapse.
- Consumer monitoring devices can detect vitals but do not actively guide intervention.

Core gap: There is a practical gap between collapse and effective first intervention.

3 Product Vision and Value Proposition

3.1 Vision

Enable any bystander to become an effective CPR responder within seconds of deployment.

3.2 Value Proposition

PulsePatch provides:

- **Zero-friction deployment**: peel-and-stick operation.
- **Immediate guidance**: prompts and rhythm coaching without app setup.
- **Real-time feedback**: continuous correction toward guideline compression pace.
- **Portability and readiness**: lightweight form factor for broad distribution.

4 Target Users and Use Environments

4.1 Primary Users

- Untrained bystanders
- Workplace safety personnel
- Gym staff and trainers
- School staff

4.2 Secondary Users

- First responders
- Ambulance crews in pre-AED phase
- Industrial safety teams

4.3 Initial Deployment Environments

- Schools and universities
- Fitness centers and sports facilities
- Offices, factories, and campuses
- Public venues where AED access may be delayed

5 Primary Use Case Flow

1. Individual collapses and is non-responsive.
2. Bystander retrieves PulsePatch and applies to sternum/chest region.
3. Device auto-activates on peel/application.
4. Device performs signal acquisition and prompt sequence (e.g., “Check responsiveness”).
5. If no response condition persists, system transitions into CPR coaching mode.
6. Device provides metronome, visual cues, and compression feedback loop.
7. Guidance continues until EMS arrival, AED takeover, or manual stop.

6 Functional Requirements

7 Non-Functional Requirements

8 Hardware and Software Architecture

8.1 Hardware Baseline (MVP Direction)

- MCU: ESP32 or nRF52 class
- ECG Front-End: AD8232 class or equivalent
- IMU: MPU6050/ICM class sensor
- Output: piezo buzzer or micro-speaker, LED indicators, optional micro-display
- Power: sealed or rechargeable Li-Po option

8.2 Software Modules

- Signal pre-processing (ECG filtering, artifact suppression)
- Motion classification and compression event detection
- State machine and transition logic
- Feedback synthesis (tempo, corrective cues, timers)
- Optional BLE data transport layer (future)

ID	Feature	Requirement	Priority
FR-01	Activation	Device shall support single-step peel-and-stick activation with no app dependency in MVP.	Must
FR-02	Placement Guidance	Device shall indicate valid placement zone and confirm start state via light/audio cue.	Must
FR-03	Sensing	Device shall capture ECG (2–3 lead equivalent) and IMU motion data for signal analysis.	Must
FR-04	State Engine	Device shall implement state machine: Idle, Activated, Acquisition, Prompt, CPR Mode, Feedback Loop.	Must
FR-05	CPR Detection	Device shall detect compression rhythm and consistency from IMU stream.	Must
FR-06	Real-time Feedback	Device shall provide corrective cues for pace too fast/slow and compression not detected.	Must
FR-07	Coaching Output	Device shall provide metronome guidance within 100–120 BPM target range.	Must
FR-08	Cycle Management	Device shall run 2-minute compression intervals with reassessment prompt.	Should
FR-09	Minimal UI	Device shall use LED + buzzer/speaker indicators with minimal cognitive load.	Must
FR-10	Manual Override	Device should support optional emergency button to force coaching mode.	Could
FR-11	Connectivity	Device may support BLE logging/app handoff in future releases.	Could
FR-12	Data Export	Future app shall support event log export for EMS handoff (non-MVP).	Could

ID	Category	Requirement	Priority
NFR-01	Activation Speed	Activation-to-guidance readiness target: < 2 seconds under standard conditions.	Must
NFR-02	Detection Accuracy	Compression detection accuracy target: $> 90\%$ in validation scenarios.	Must
NFR-03	Rhythm Accuracy	Compression BPM estimation error target: ± 5 BPM.	Must
NFR-04	Physical Constraints	Form factor target: < 15 mm thickness, < 80 g mass.	Should
NFR-05	Power	Standby life target: > 6 months for deployable readiness configuration.	Should
NFR-06	Usability	Interface shall be understandable in high-stress conditions with minimal training.	Must
NFR-07	Safety Labeling	Product messaging shall clearly avoid diagnostic/defibrillation claims in MVP.	Must

9 Safety, Regulatory, and Claims Strategy

- PulsePatch is designed as a response guidance aid, not a diagnostic monitor.
- Anticipated regulatory pathway likely aligns with medical device controls (e.g., Class II context depending on jurisdiction and claims).
- Claims language must be tightly scoped to avoid unsupported clinical assertions before validation.
- Clinical and mannequin-based studies required before broad-scale launch claims.

10 Success Metrics (KPIs)

10.1 Product KPIs

- Activation time: < 2 seconds
- Compression detection accuracy: $> 90\%$
- Compression BPM estimation accuracy: within ± 5 BPM
- Readiness/battery standby: > 6 months target

10.2 User and Outcome KPIs

- Reduced time to first compression in simulation studies
- Improved CPR rhythm adherence versus no-guidance baseline
- Increased bystander confidence scores (survey instrument)

11 Roadmap

Phase	Timeframe	Objectives
Prototype	0–3 months	IMU-based detection, metronome + LED feedback, first physical patch form factor.
Functional MVP	3–6 months	ECG integration, robust state logic, improved enclosure and UX.
Pilot Testing	6–12 months	Mannequin protocols, controlled user studies, iteration on feedback effectiveness.
Regulatory Path	12+ months	Clinical planning, submission package preparation, certification pathway execution.

12 Business Model and Commercialization

Revenue Streams

- Direct unit sales
- Bulk B2B procurement (schools, gyms, workplaces, industrial facilities)
- Future software/app subscription for event analytics and response records

Initial Pricing Direction

- Target unit range: USD \$40–\$120 depending on model and feature set.

13 Risks and Mitigations

Risk Area	Key Risk	Mitigation Approach
Signal Quality	Motion artifact and low perfusion reduce signal reliability.	Multi-sensor fusion, robust filtering, confidence thresholds, fallback logic.
Algorithmic Error	False positives/negatives in event interpretation.	Conservative trigger logic, simulation data expansion, staged validation.
UX Under Stress	User confusion during emergency conditions.	Minimal interface, plain prompts, rapid usability testing loops.
Regulatory	Delays due to claim scope or evidence gaps.	Early regulatory advisory, claim discipline, planned validation milestones.

14 Open Decisions (For Next Revision)

- Single-use versus limited-reuse hardware strategy
- Final enclosure geometry and adhesive material standard
- Voice prompt inclusion in MVP versus premium release
- Data retention and privacy policy for connected features

15 Conclusion

PulsePatch addresses an urgent intervention gap in cardiac emergencies by combining zero-friction deployment with active CPR coaching. With focused prototyping, rigorous validation, and disciplined regulatory strategy, PulsePatch has the potential to become a practical, scalable emergency response product for public and institutional environments.

Prepared for presentation, partner outreach, and early investment conversations.